

Logistic Regression Assumptions and Diagnostics in R

 sthda.com/english/articles/36-classification-methods-essentials/148-logistic-regression-assumptions-and-diagnostics-in-r

kassambara

March 11, 2018



The **logistic regression** model makes several **assumptions** about the data.

This chapter describes the major assumptions and provides practical guide, in R, to check whether these assumptions hold true for your data, which is essential to build a good model.

Make sure you have read the logistic regression essentials in Chapter [@ref\(logistic-regression\)](#).

Logistic regression assumptions

The logistic regression method assumes that:

- The outcome is a binary or dichotomous variable like yes vs no, positive vs negative, 1 vs 0.
- There is a linear relationship between the logit of the outcome and each predictor variables. Recall that the logit function is $\text{logit}(p) = \log(p/(1-p))$, where p is the probabilities of the outcome (see Chapter [@ref\(logistic-regression\)](#)).
- There is no influential values (extreme values or outliers) in the continuous predictors
- There is no high intercorrelations (i.e. multicollinearity) among the predictors.

To improve the accuracy of your model, you should make sure that these assumptions hold true for your data. In the following sections, we'll describe how to diagnostic potential problems in the data.

Loading required R packages

- **tidyverse** for easy data manipulation and visualization
- **broom** : creates a tidy data frame from statistical test results

```
library(tidyverse)
library(broom)
theme_set(theme_classic())
```

Building a logistic regression model

We start by computing an example of logistic regression model using the `PimaIndiansDiabetes2` [mlbench package], introduced in Chapter @ref(classification-in-r), for predicting the probability of diabetes test positivity based on clinical variables.

```
data("PimaIndiansDiabetes2", package = "mlbench")
PimaIndiansDiabetes2 <- na.omit(PimaIndiansDiabetes2)

model <- glm(diabetes ~., data = PimaIndiansDiabetes2,
             family = binomial)

probabilities <- predict(model, type = "response")
predicted.classes <- ifelse(probabilities > 0.5, "pos", "neg")
head(predicted.classes)

##    4    5    7    9   14   15
## "neg" "pos" "neg" "pos" "pos" "pos"
```

Logistic regression diagnostics

Linearity assumption

Here, we'll check the linear relationship between continuous predictor variables and the logit of the outcome. This can be done by visually inspecting the scatter plot between each predictor and the logit values.

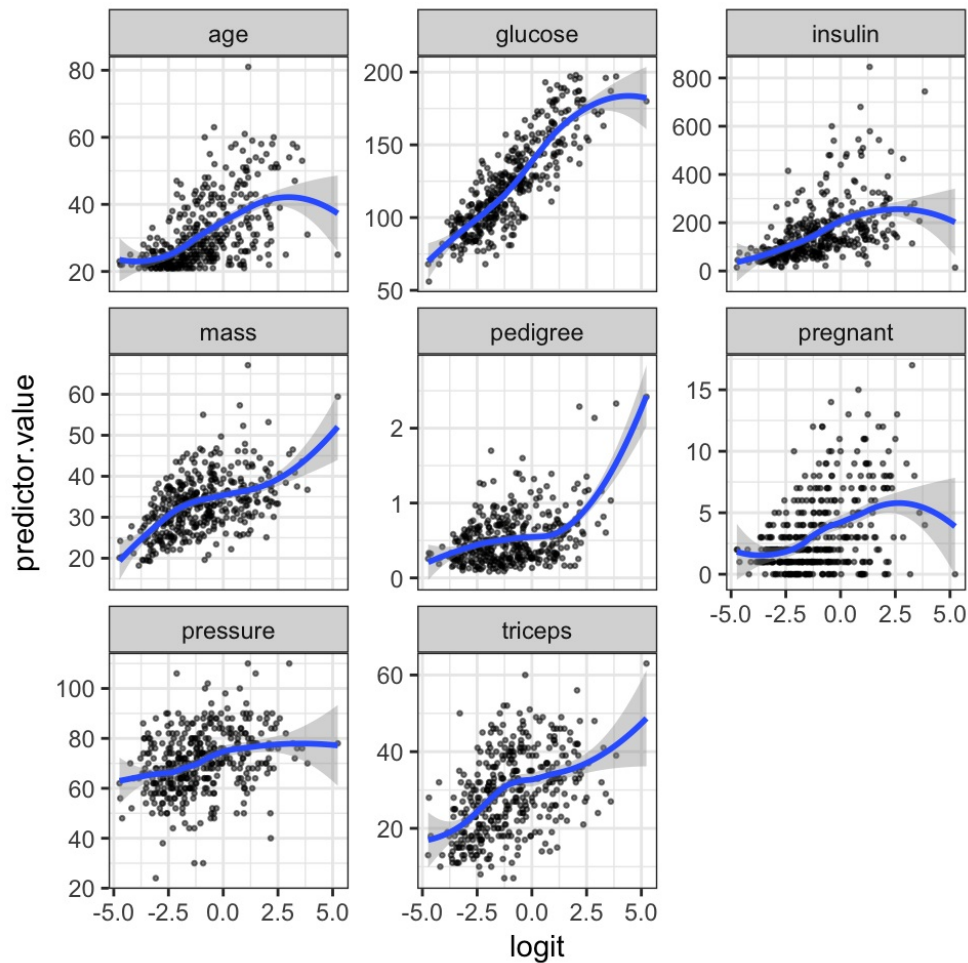
1. Remove qualitative variables from the original data frame and bind the logit values to the data:

```
mydata <- PimaIndiansDiabetes2 %>%
  dplyr::select_if(is.numeric)
predictors <- colnames(mydata)

mydata <- mydata %>%
  mutate(logit = log(probabilities/(1-probabilities))) %>%
  gather(key = "predictors", value = "predictor.value", -logit)
```

2. Create the scatter plots:

```
ggplot(mydata, aes(logit, predictor.value)) +
  geom_point(size = 0.5, alpha = 0.5) +
  geom_smooth(method = "loess") +
  theme_bw() +
  facet_wrap(~predictors, scales = "free_y")
```



The smoothed scatter plots show that variables glucose, mass, pregnant, pressure and triceps are all quite linearly associated with the diabetes outcome in logit scale.

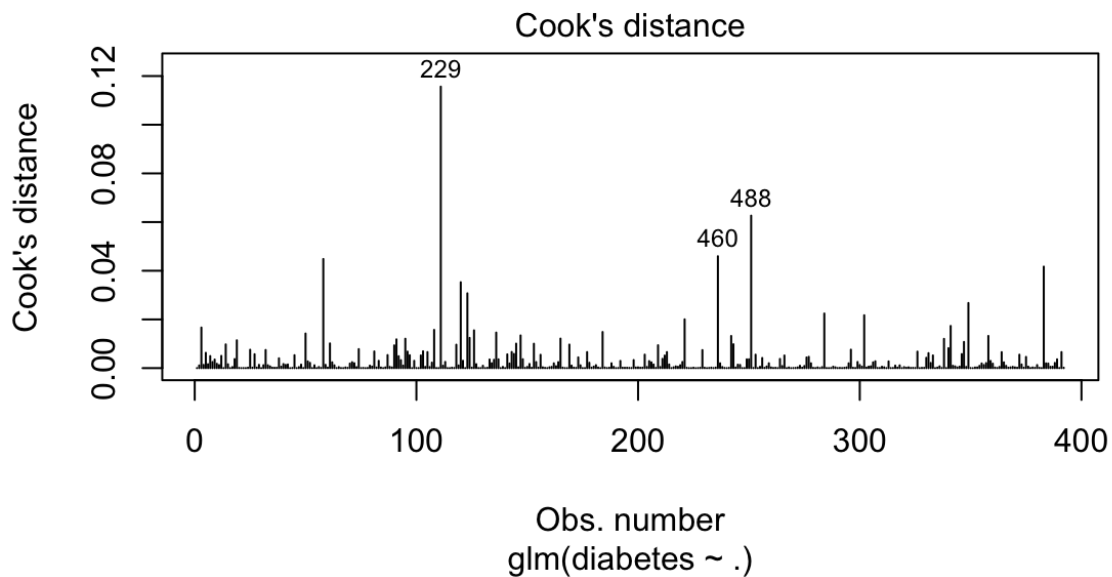
The variable age and pedigree is not linear and might need some transformations. If the scatter plot shows non-linearity, you need other methods to build the model such as including 2 or 3-power terms, fractional polynomials and spline function (Chapter [@ref\(polynomial-and-spline-regression\)](#)).

Influential values

Influential values are extreme individual data points that can alter the quality of the logistic regression model.

The most extreme values in the data can be examined by visualizing the Cook's distance values. Here we label the top 3 largest values:

```
plot(model, which = 4, id.n = 3)
```



Note that, not all outliers are influential observations. To check whether the data contains potential influential observations, the standardized residual error can be inspected. Data points with an absolute standardized residuals above 3 represent possible outliers and may deserve closer attention.

The following R code computes the standardized residuals (`.std.resid`) and the Cook's distance (`.cooksd`) using the R function `augment()` [broom package].

```
model.data <- augment(model) %>%  
  mutate(index = 1:n())
```

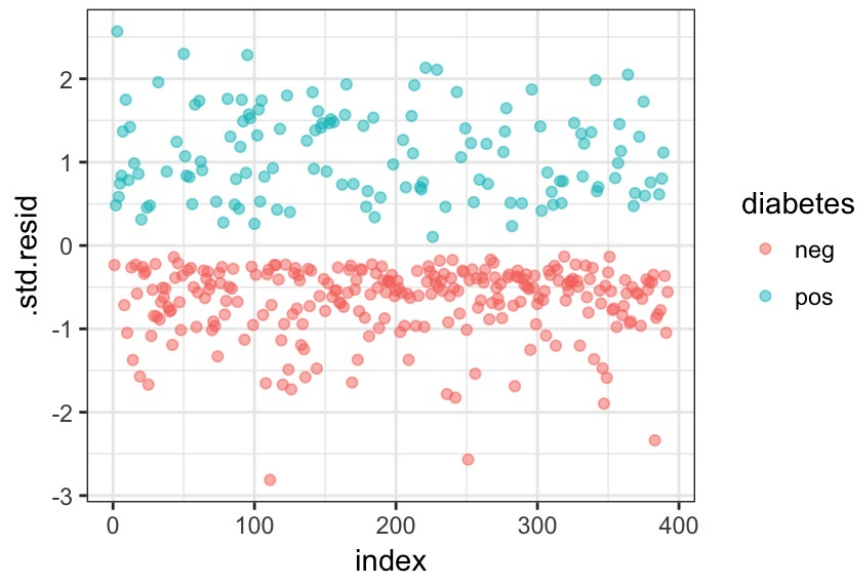


The data for the top 3 largest values, according to the Cook's distance, can be displayed as follow:

```
model.data %>% top_n(3, .cooksd)
```

Plot the standardized residuals:

```
ggplot(model.data, aes(index, .std.resid)) +  
  geom_point(aes(color = diabetes), alpha = .5) +  
  theme_bw()
```



Filter potential influential data points with `abs(.std.res) > 3` :

```
model.data %>%
  filter(abs(.std.resid) > 3)
```

There are no influential observations in our data.

When you have outliers in a continuous predictor, potential solutions include:

- Removing the concerned records
- Transform the data into log scale
- Use non parametric methods

Multicollinearity

Multicollinearity corresponds to a situation where the data contain highly correlated predictor variables. Read more in Chapter [@ref\(multicollinearity\)](#).

Multicollinearity is an important issue in regression analysis and should be fixed by removing the concerned variables. It can be assessed using the R function `vif()` [car package], which computes the variance inflation factors:

```
car::vif(model)

## pregnant glucose pressure triceps insulin mass pedigree age
## 1.89 1.38 1.19 1.64 1.38 1.83 1.03 1.97
```

As a rule of thumb, a VIF value that exceeds 5 or 10 indicates a problematic amount of collinearity. In our example, there is no collinearity: all variables have a value of VIF well below 5.

Discussion

This chapter describes the main assumptions of logistic regression model and provides examples of R code to diagnostic potential problems in the data, including non linearity between the predictor variables and the logit of the outcome, the presence of influential observations in the data and multicollinearity among predictors.

Fixing these potential problems might improve considerably the goodness of the model. See also, additional performance metrics to check the validity of your model are described in the Chapter @ref(classification-model-evaluation).

2 Notes

Enjoyed this article? Give us 5 stars (just above this text block)! Reader needs to be STHDA member for voting. I'd be very grateful if you'd help it spread by emailing it to a friend, or sharing it on Twitter, Facebook or Linked In.

Show me some love with the like buttons below... Thank you and please don't forget to share and comment below!!

Recommended for You!

[Machine Learning Essentials: Practical Guide in R](#)

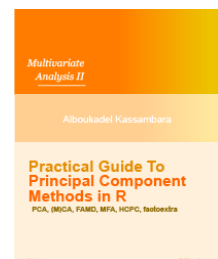
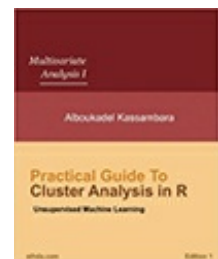
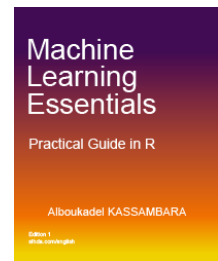
[Practical Guide to Cluster Analysis in R](#)

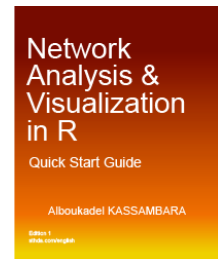
[Practical Guide to Principal Component Methods in R](#)

[R Graphics Essentials for Great Data Visualization](#)

[Network Analysis and Visualization in R](#)

[More books on R and data science](#)





Recommended for you

This section contains best data science and self-development resources to help you on your path.

Coursera - Online Courses and Specialization

Data science

Popular Courses Launched in 2020

Trending Courses

Books - Data Science

Our Books

Others

You are not authorized to post a comment

Michael U

Visitor

#682 02/01/2019 at 12h17

Dear Alboukadel Kassambara,

first of all thank you very much for this - any many other - useful post. I like your articles a lot because they really focus on the essentials.

I have one question regarding the linearity assumption check. You write that

Quotation :

The smoothed scatter plots show that variables glucose, mass, pregnant, pressure and triceps are all quite linearly associated with the diabetes outcome in logit scale.

The variable age and pedigree is not linear and might need some transformations.



Could you please explain briefly **how** one can see that?

The scatter plot and smoother for the variable *glucose* shows, without any doubt, a very clear linear relationship. Also, I agree that *mass*, *triceps* and even *pressure* show linear relations. However, I cannot see it for the remaining variables. Most importantly, to my eye variables *age* and *pregnant* look very similar in terms of the smoothed line. But **why** is then *pregnant* linearly associated with the logit, while *age* is not? Is this because for *age*, the data points are somewhat clumped together in the lower left part of the plot, while for *pregnant* the data points are more evenly spread over the range of the plot?

Similarly, on what grounds can one state that *pedigree* is not linear, in other words, what visual hints are there in the scatter plot for *pedigree*?

I understand that the visual inspection of plots is not an exact science and that it takes some experience. Since I still lack this experience, I will highly appreciate your helpful comment on this matter.

Thank you very much in advance.

Best wishes,
Michael